

How to Be a Good AI Researcher

Tao LIN

September 22, 2026



1 Course Logistics

2 Recitation

3 How to Do Research

- The Illustrated Guide to a Ph.D.
- 10 Easy Ways to Fail a Ph.D.
- Introduction to the Research Workflow (a CV Perspective)
- How to Make Steady Progress?
- Research Efficiency in 2026
- How to Keep Track With the Literature?
- How to Read Papers?

4 Concluding Remarks

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Course Schedule

Week	Date	Topics
1	2026. Sep. 01	How to communicate
2	2026. Sep. 08	How to do presentation
3	2026. Sep. 15	How to be a good AI researcher (I): doing research I
4	2026. Sep. 22	How to be a good AI researcher (II): productivity and career
5	2026. Sep. 29	How to be a good AI researcher (III): academic paper writing and peer reviews
6	2026. Oct. 13	Sharing the experience of writing excellent academic papers and rebuttal
7	2026. Oct. 20	Practice course I
8	2026. Oct. 27	Practice course II

Acknowledgement

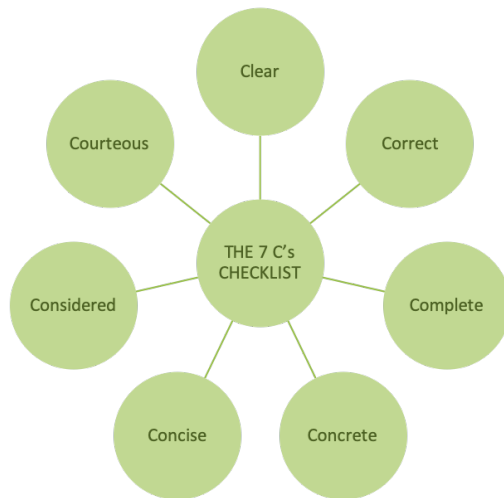
- The 7 Cs of Communication, World of Work Project
- Awesome Tips, JiaBin Huang
- The illustrated guide to a Ph.D., Matt Might
- 10 easy ways to fail a Ph.D., Matt Might
- GAMES003

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In the previous two weeks...

We learn how to communicate: the 7 **C**'s of communication



Please revisit our previous lecture when necessary.

We learn the principles of presentation

Great talks require effort & time!

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- 5 Guide, highlight, and interpret the key results for your audience!
- 6 Practice almost makes perfect. Rehearse your presentation from the aspect of your audience!

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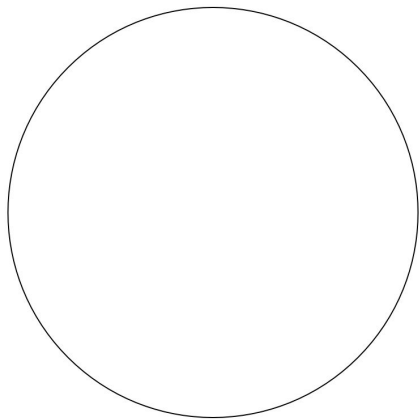
What a Ph.D. is?

¹ <https://matt.might.net/articles/phd-school-in-pictures/>

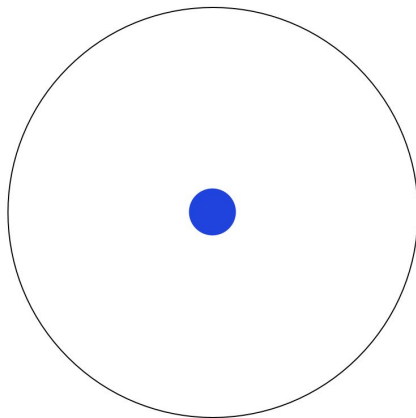
What a Ph.D. is?

Let's get started with a famous example¹

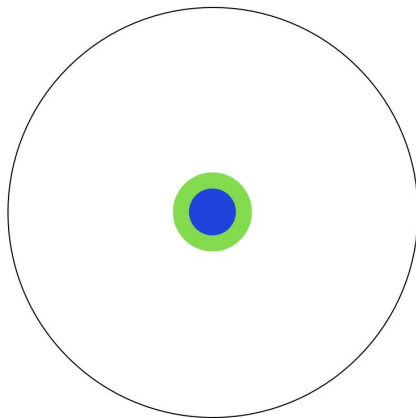
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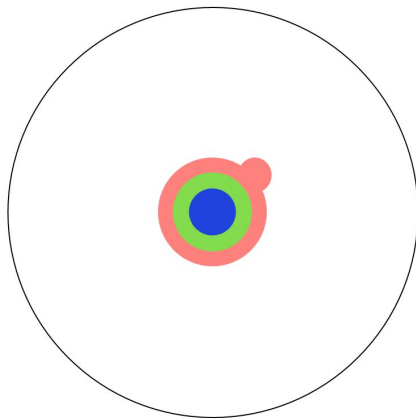
Imagine a circle that contains all of human knowledge



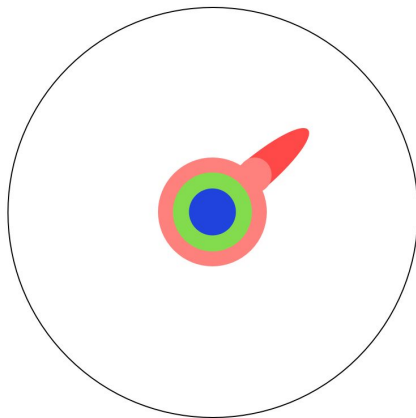
By the time you finish elementary school, you know a little



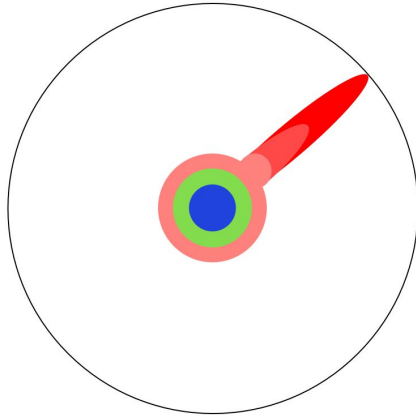
By the time you finish high school, you know a bit more



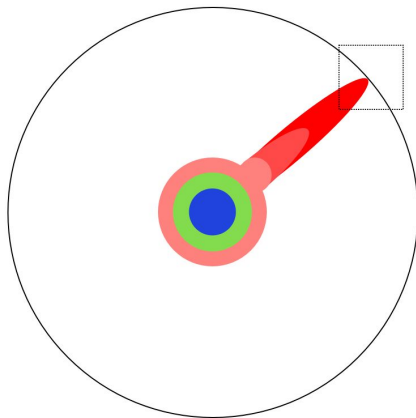
With a bachelor's degree, you gain a specialty



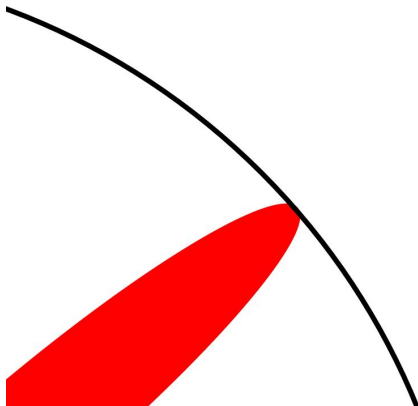
A master's degree deepens that specialty



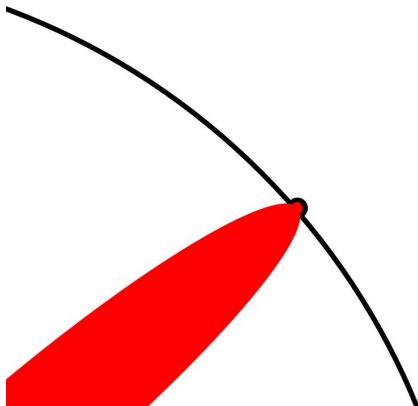
Reading research papers takes you to the edge of human knowledge



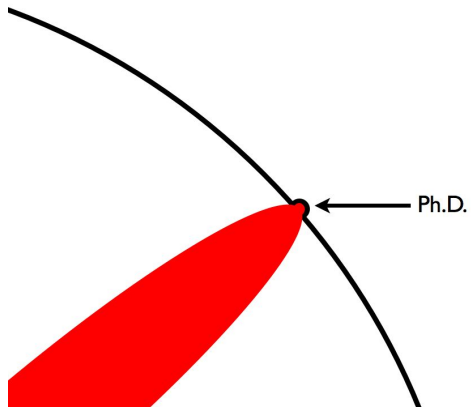
Once you're at the boundary, you focus



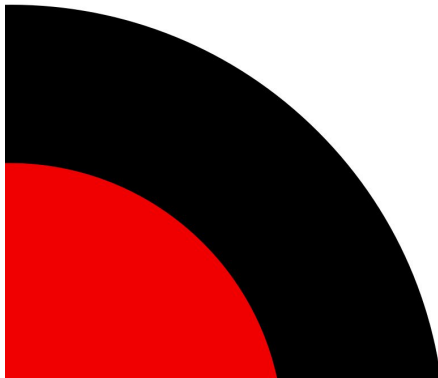
You push at the boundary for a few years



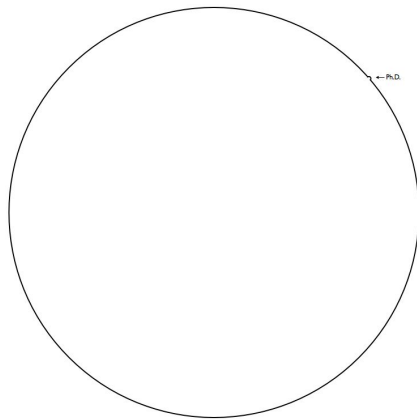
Until one day, the boundary gives way



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Of course, the world looks different to you now



Of course do not forget the bigger picture

Back to something concrete

Let's talk about 10 easy ways to fail a Ph.D.

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Focus on grades or coursework

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$$\text{Optimal GPA} = \text{Minimum Required GPA} + \epsilon$$

Learn too much

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It is important to trade off learning width and learning depth.

Expect perfection

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- it is important to get started, and improve it through an iterative process.

Procrastinate



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We will have a separate session on tips for boosting productivity.

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- student should know more than the advisor about the topic
- once it happens, the student should start choosing the topics

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- You need much larger levels of independence

Aim too low

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Aiming low leaves no room for uncertainty. And research is always uncertain.

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- It is the training that counts—not the topic.

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- When you have a few good publications and can convince your committee → thesis proposal
- Once you can repeatedly conduct research to meet the standards of peer review → defense
- Once you can draw a unifying theme → thesis and dissertation

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- He ignored that warning.
- He was lucky: it added only one more semester to his Ph.D.

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How to Make Steady Progress?



Mentor/Advisor



Research Skills

How to Make Steady Progress?



Mentor/Advisor

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Mentor/Advisor

Please check our week2 slides on
“How to work with your advisor effectively”

How to Make Steady Progress?



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Key messages:

How to Make Steady Progress?



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Key messages:

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How to Make Steady Progress?



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Key messages:

- Your advisor is an input-output machine
- Show your work (what, why, how, etc)

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Key messages:

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- Provide contexts (maintain a detailed meeting minute and start with why)

How to Make Steady Progress?



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Key messages:

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- Present failures
- Provide contexts (maintain a detailed meeting minute and start with why)
- Set expectation (I will do XXX by YYY)

How to Make Steady Progress?



Mentor/Advisor



Research Skills

How to Make Steady Progress?



Research Skills

How to Make Steady Progress? Imagine success



How to Make Steady Progress? Work backward

- Forward: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$
- Backward: $D \rightarrow E$; $C \rightarrow D$; $B \rightarrow C$; $A \rightarrow B$



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Why?

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- See upper bound early

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- See the final outcome early
- See upper bound early
- Focus on each task with perfect inputs

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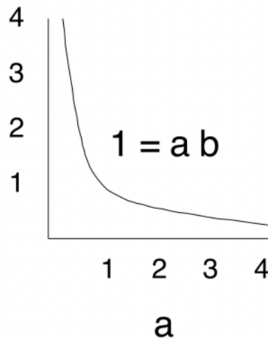


- See the final outcome early
- See upper bound early
- Focus on each task with perfect inputs
- Understand what is needed

How to Make Steady Progress? Use toy example to capture the main idea.



Color constancy problem



How to Make Steady Progress? Baselines first

Don't know where to start?



Let's build a sheep counter!

How to Make Steady Progress? Baselines first

Don't know where to start?



Let's build a sheep counter!

- Start with trying out baseline methods.

How to Make Steady Progress? Baselines first

Don't know where to start?



Let's build a sheep counter!

- Start with trying out baseline methods.
- It helps identify the limitations of SOTA.

How to Make Steady Progress? Baselines first

Don't know where to start?



Let's build a sheep counter!

- Start with trying out baseline methods.
- It helps identify the limitations of SOTA.
- If they work perfectly well, then??

How to Make Steady Progress? Simple case first



If your method does not work on simple/trivial cases,

How to Make Steady Progress? Simple case first



If your method does not work on simple/trivial cases,
how could you expect it to work on unconstrained, real-world cases?

How to Make Steady Progress? One thing at a time

`batch_size = 4 → batch_size = 16`

`learning_rate = 1e-3 → learning_rate = 1e-5`

`use_abc_loss = False → use_abc_loss = True`

When doing experiments, change exactly ONE thing at a time.

How to Make Steady Progress? One thing at a time

`batch_size = 4 → batch_size = 16`

`learning_rate = 1e-3 → learning_rate = 1e-5`

`use_abc_loss = False → use_abc_loss = True`

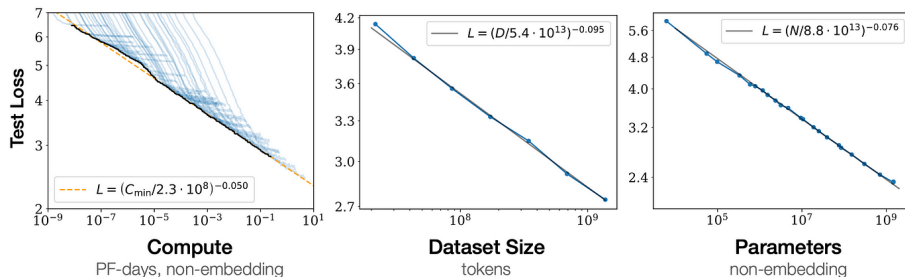
When doing experiments, change exactly ONE thing at a time.
→ This helps you understand what the results mean.

How to Make Steady Progress? Identify proxy

- Do not use full-scale experiments (that may take weeks to complete) to validate your ideas

²Scaling Laws for Neural Language Models, <https://arxiv.org/abs/2001.08361>

How to Make Steady Progress? Identify proxy

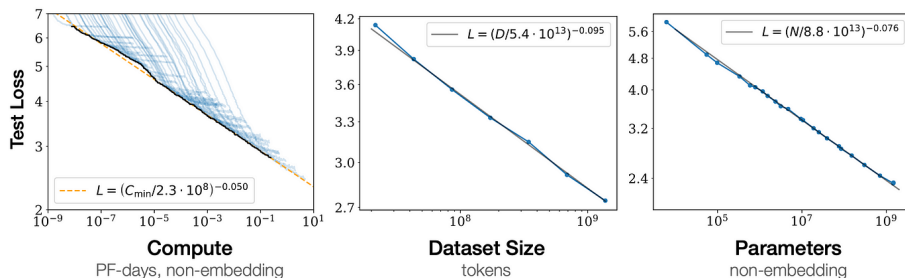


Example: *scaling rule* in Deep Learning²

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²Scaling Laws for Neural Language Models, <https://arxiv.org/abs/2001.08361>

How to Make Steady Progress? Identify proxy



Example: *scaling rule* in Deep Learning²

- Do not use full-scale experiments (that may take weeks to complete) to validate your ideas
- Run smaller-scale/simpler experiments with short turnaround time
→ you get to iteratively refine your ideas a lot faster.

²Scaling Laws for Neural Language Models, <https://arxiv.org/abs/2001.08361>

How to Make Steady Progress? Automate everything / make the best use of machine time

If you find that you need to do the same task twice, write a script for that.

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Plan your experiments → your machines work for you while you are not working.

How to Make Steady Progress? Automate everything / make the best use of machine time

If you find that you need to do the same task twice, write a script for that.

Plan your experiments → your machines work for you while you are not working.

Your future self will thank you.

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- How to Keep Track With the Literature?
- How to Read Papers?

4 Concluding Remarks

Research Efficiency in 2026: Shorten the Feedback Loop

Decide → Design → Build → Run → Analyze → Verify → Decide
again

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- Use AI and automation to reduce waiting between these steps.

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- Use AI and automation to reduce waiting between these steps.
- Keep problem choice, scientific judgment, and final verification with the researcher.

Research Efficiency in 2026: Shorten the Feedback Loop

Decide → Design → Build → Run → Analyze → Verify → Decide
again

- Use AI and automation to reduce waiting between these steps.
- Keep problem choice, scientific judgment, and final verification with the researcher.
- Optimize for **trustworthy information gained per unit cost**, not activity completed.

Cognitive Efficiency: Let Agents Map; You Verify

- 1 State the research question and the decision the review must support.

Cognitive Efficiency: Let Agents Map; You Verify

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- 4 Save a small claim–source table that you can update and audit later.

An AI summary is a navigation aid, not evidence.

Engineering Efficiency: Delegate Bounded Artifacts

Good tasks for agents

- Search and organize candidates

The researcher retains

- Problem framing and priorities

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- Scaffold code, tests, and scripts

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The researcher retains

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- Assumptions and evaluation design
- Conflict resolution and integration
- Evidence checks and scientific claims

Parallel agents need independent scopes; one integrator owns the conflicts.

Computational Efficiency: Scale Only After Cheap Evidence

Sanity check → Tiny run → Baseline → Validated proxy → Staged scale-up

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- Validate that the cheap proxy preserves the important failures and method ranking.
- Set budgets and stopping rules before a large run starts.
- Track **attention**, **wall-clock time**, and **compute** separately.

A Practical 2026 Agent Brief

Task	One bounded objective with an explicit owner
Context	Relevant files, papers, prior results, and dependencies
Constraints	Time, compute, methods, style, and non-goals
Deliverable	Concrete artifact: patch, table, plot, or evidence memo
Success check	Test, metric, comparison, or acceptance criterion
Permissions	What may be read, changed, run, or contacted
Stop condition	When to return, escalate, or declare uncertainty

“Do the research” is not a task specification.

Fast Is Not Correct: Verify Before You Trust

Three checks

- 1 Does the source exist?

A 2026 snapshot

Fast Is Not Correct: Verify Before You Trust

Three checks

- 1 Does the source exist?
- 2 Does it support the claim?

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Fast Is Not Correct: Verify Before You Trust

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- Best SWE-bench Science pass@1: below 50%.

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- Best SWE-bench Science pass@1: below 50%.
- Best AutoMat setting reported: 53%.

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Primary sources (version-pinned): [SWE-bench Science v2](#); [AutoMat v2](#); [citation reliability v1](#).
Results are task- and configuration-specific; treat them as calibration, not universal limits.

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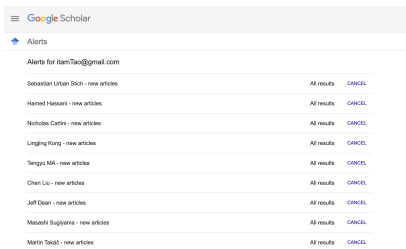
Machine Learning Arxiv Papers per Year



From Jeff Dean's Twitter (Year 2018).

What should we do?

Track the people, not the papers



Google Scholar Alerts for itemTao@gmail.com

Sebastian Urban Siddh - new articles	All results	CANCEL
Hamed Hassani - new articles	All results	CANCEL
Nicholas Carlini - new articles	All results	CANCEL
Lingling Kong - new articles	All results	CANCEL
Tengyu MA - new articles	All results	CANCEL
Chen Liu - new articles	All results	CANCEL
Jeff Dean - new articles	All results	CANCEL
Masashi Sugiyama - new articles	All results	CANCEL
Martin Takáč - new articles	All results	CANCEL

- There are far fewer key people *who are driving the field forward* than the number of papers
- Check out who the authors are when you read papers.
- Subscribe to alerts for the people and topics central to your question.
- Over time you will recognize the important ones.

Read papers with good related work

A good related work section saves you so much time by providing a clear, organized view of prior work.

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Side note: please save others' time by writing a good related work.

Organize the papers

				
<u>BibTeX</u>	EndNote	Mendeley	Papers	Zotero
Free license	Clarivate (commercial)	Elsevier (commercial)	Digital Science Research & Solutions Inc. (commercial)	Corporation for Digital Scholarship (non profit)
Free of charge	274,95 USD per year (149,495 USD for students)	Freemium	60 USD per year (36 USD for students)	Free of charge
Windows macOS Linux	Windows macOS	Windows macOS Linux	Windows macOS	Windows macOS Linux
Compatible with: (MS Word) Google Docs (OpenOffice) (LibreOffice) LaTeX	Compatible with: MS Word (Google Docs) OpenOffice LibreOffice (LaTeX)	Compatible with: MS Word (Google Docs) OpenOffice LibreOffice (LaTeX)	Compatible with: MS Word Google Docs OpenOffice LibreOffice (LaTeX)	Compatible with: MS Word Google Docs OpenOffice LibreOffice LaTeX
-	7k+ citation styles	9k+ citation styles	9k+ citation styles	10k+ citation styles

Updated on 17/04/2023

Consider to use a reference manager

Organize the papers

Your personal research assistant

Zotero is a free, easy-to-use tool to help you collect, organize, annotate, cite, and share research.

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The screenshot shows the Zotero application window. On the left is a sidebar with a tree view containing 'My Library' and several subfolders like 'Book Reviews', 'Colonial Medicine', 'Dissertation', 'Science and Empire', 'Teaching', 'Mapping', 'Open Access', 'Text Mining', and 'Visualization'. The main pane displays a list of research papers with columns for Title, Creator, and Year. The paper 'Circulation of Medicine in the Early Modern Atlantic World' by Cook and Walker (2013) is selected. On the right, a details pane shows information about the selected item, including its type (Journal Article), title, authors (Cook, Harold J. and Walker, Timothy D.), abstract, publication (Social History of Medicine), and volume (26).

Title	Creator	Year
Guerre, maladie, empire. Les services de santé militaires en ...	Zaug	2016
Officiers de santé et soignantes créoles face à la fièvre jaune	Nobl	2016
The Emergence of Tropical Medicine in France	Osborne	2014
Colonial Disease, Translation, and Enlightenment: Franco-Brit...	Charters	2014
Trading in Drugs through Philadelphia in the Eighteenth Centu...	Wilson	2013
The Medicines Trade in the Portuguese Atlantic World: Acquisi...	Walker	2013
Leprosy and Slavery in Suriname: Godfried Schilling and the Fr...	Snelders	2013
Medical Experimentation and Race in the Eighteenth-century ...	Schleibinger	2013
The Circulation of Bodily Knowledge in the Seventeenth-centu...	Gómez	2013
Circulation of Medicine in the Early Modern Atlantic World	Cook and Walker	2013

Item Type: Journal Article
 Title: Circulation of Medicine in the Early Modern Atlantic World
 Author: Cook, Harold J.
 Author: Walker, Timothy D.
 Abstract: The search for powerful drugs has caused people and commodities to move around the globe for many centuries, as it still does...
 Publication: Social History of Medicine
 Volume: 26

I use Zotero in my case

Triage before reading the full paper

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Use accessible resources to decide where a careful reading will be most valuable.

Triage before reading the full paper

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- 2021-22 Allen School Distinguished Lecture Series** (Updated 3 days ago) View full playlist
- 2021-22 Allen School Colloquium** (Updated 3 days ago) View full playlist
- 2021 Information-Theoretic Cryptography (ITC) Conference** (Updated 3 days ago) View full playlist
- Research Seminar: Mobile & Wireless Systems** (Updated 3 days ago) View full playlist
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Use accessible resources to decide where a careful reading will be most valuable.

- a talk, a YouTube video, teaser results, an introductory video, or an overview figure.

Read with a purpose

Before investing time in reading a paper, think about WHY you are reading it.

You almost never need to read a paper from top to bottom.

Read with a purpose

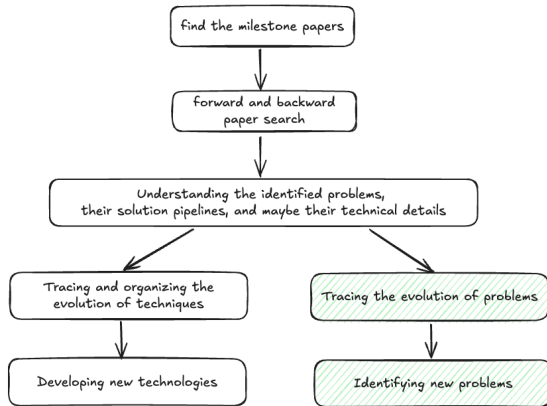
Before investing time in reading a paper, think about WHY you are reading it.

Are you reading for

- the experimental setup?
- the organization?
- the story?
- the style?
- the method?
- or the visualization?

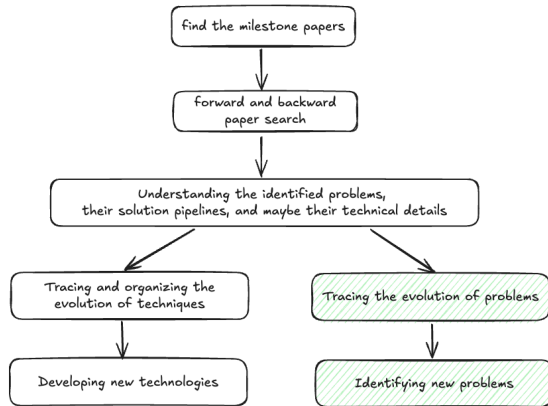
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How to be a domain expert by reading papers



How to identify milestone papers

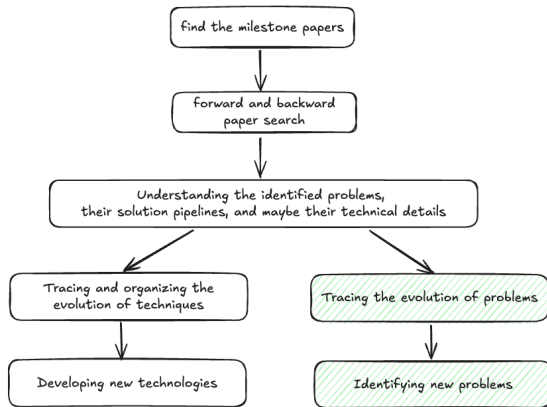
How to be a domain expert by reading papers



How to identify milestone papers

- Find the most cited papers in this field and sort them by date.

How to be a domain expert by reading papers

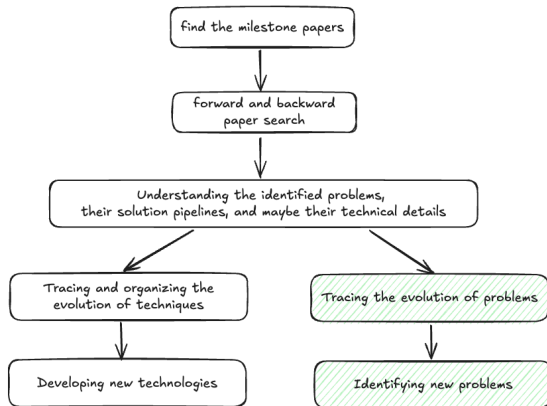


How to identify milestone papers

- Find the most cited papers in this field and sort them by date.
- Seek advice from experienced researchers.

How to be a domain expert by reading papers

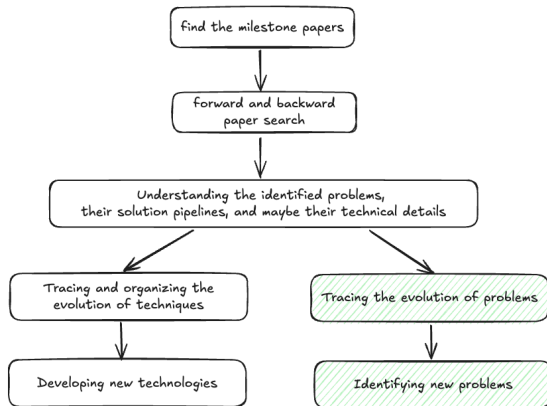
How to find papers between two milestone papers



How to be a domain expert by reading papers

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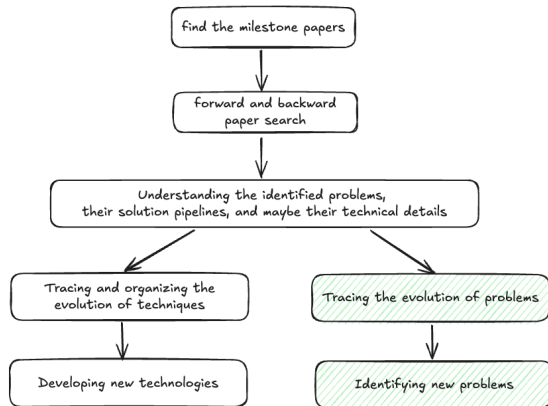
- Forward pass



How to be a domain expert by reading papers

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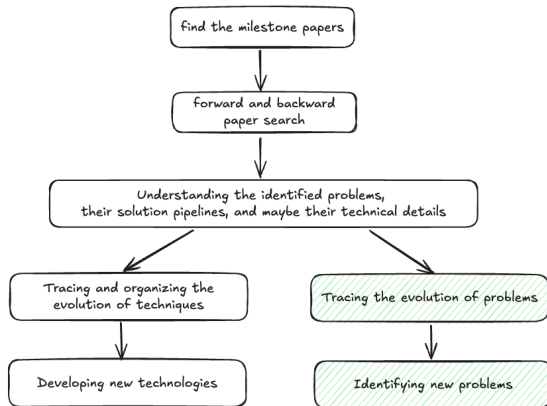
- Forward pass
 - papers cited by the milestone paper
 - foundation work



How to be a domain expert by reading papers

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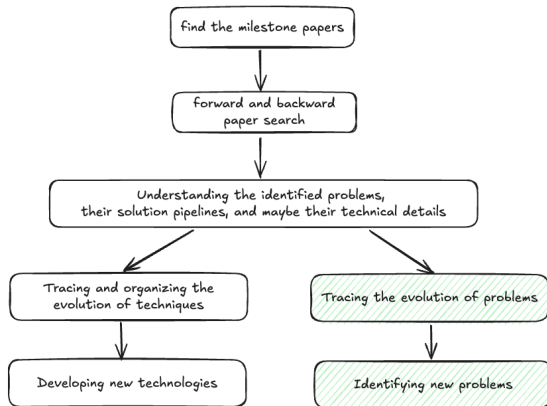
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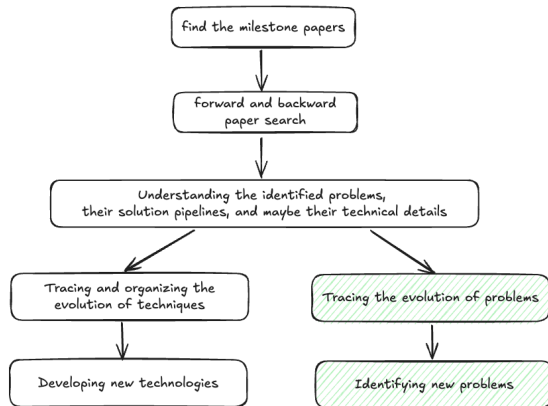
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How to find papers between two milestone papers

- Forward pass
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 - foundation work
- Backward pass
 - papers cite the milestone paper
 - followup work

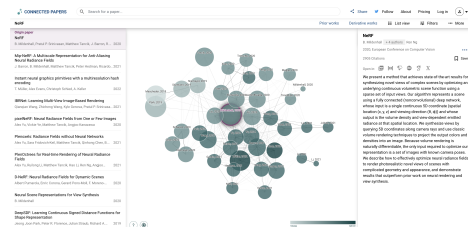


How to be a domain expert by reading papers



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Connected Papers

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Template: When reading a paper, please try to answer these questions (using 1-3 sentences).

- What is the problem? / How important is it?

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- What is the problem? / How important is it?
- What are the insights?

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- What is the solution? / Is it feasible?

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- What is the takeaway message?

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- What is the problem? / How important is it?
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- What is the takeaway message?
- Will this paper win the test of time award?

Template: When reading a paper, please try to answer these questions (using 1-3 sentences).

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- What are the insights?
- What is the solution? / Is it feasible?
- What is the takeaway message?
- Will this paper win the test of time award?
- Name one reason why this paper should have not appeared in NIPS, ICML, ICLR, etc?

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Accelerate execution. Preserve
judgment. Verify evidence.